



# KeywordIQ

AI-Powered SEO Keyword Generator for Health & Supplement Products

MAI417-3 Deep Learning | M.Sc. AIML | NeuralHack 2026 | Christ (Deemed to be University), Bangalore

Validation AUC

**0.9668**

Multi-label AUC-ROC

Val Accuracy

**99.67%**

Binary classification

Best Epoch

**22 / 30**

EarlyStopping at 27

Parameters

**894K**

3.41 MB model size

## What Was Built

KeywordIQ is a trained deep learning system that predicts relevant SEO keywords from Amazon Health & Supplement product titles. The core use case is reseller automation — a seller with 500 products can upload a CSV and download it with a keywords column added in seconds.

## Task Formulation

Multi-Label Binary Classification: given a title, predict which of 500 SEO keywords apply. Each keyword is an independent sigmoid prediction — not a competition between classes (no softmax). Binary Cross-Entropy loss, Adam optimizer.

## Dataset

| Source          | Amazon Health & Supplements |
|-----------------|-----------------------------|
| Raw rows        | 142,238 product listings    |
| After cleaning  | 40,349 unique samples       |
| Training / Val  | 34,296 / 6,053 (85-15%)     |
| Keyword vocab   | Top-500 most frequent       |
| Sequence length | 60 tokens (padded)          |

## Regularization Results

Dropout (0.4 + 0.3) and L2 weight decay ( $1e-4$ ) prevented overfitting. At best epoch: Train AUC = 0.9631, Val AUC = 0.9668 — validation exceeds training, confirming Dropout worked correctly.

## Model Architecture

| INPUT         | Title → tokenize → pad → 60 tokens                                                 |
|---------------|------------------------------------------------------------------------------------|
| EMBEDDING     | 10,001 vocab × 64 dims   shared by both branches   640,064 params                  |
| CNN BRANCH    | Conv1D(128, k=3) → trigrams Conv1D(64, k=2) → bigrams GlobalMaxPool → 64-dim       |
| BiLSTM BRANCH | Bidirectional LSTM(64) Forward + Backward reading 128-dim context vector           |
| MERGE         | Concatenate CNN + BiLSTM 64 + 128 = 192-dim vector                                 |
| DENSE + DROP  | Dense(256) + L2 + Dropout(0.4) Dense(128) + L2 + Dropout(0.3)                      |
| OUTPUT        | Dense(500, sigmoid) 500 independent probabilities Threshold $\geq 0.30$ → keywords |

## Training Highlights

| Epoch | Tr AUC | Val AUC | Val Loss | Status                |
|-------|--------|---------|----------|-----------------------|
| 1     | 0.498  | 0.500   | 0.0275   | Starting              |
| 8     | 0.561  | 0.624   | 0.0222   | Learning begins       |
| 11    | 0.869  | 0.923   | 0.0165   | Strong generalization |
| 15    | 0.944  | 0.955   | 0.0135   | Near optimal          |
| 22 ★  | 0.963  | 0.9668  | 0.0114   | BEST — saved          |
| 27    | —      | —       | —        | EarlyStopping         |

Sample Model Predictions (Live Inference)

| Product Title                                                      | Predicted Keywords                                    | Confidence  |
|--------------------------------------------------------------------|-------------------------------------------------------|-------------|
| Optimum Nutrition Gold Standard Whey Protein Powder 5 lb           | protein powder, whey protein, muscle protein          | 96% 95% 74% |
| Nature Made Vitamin D3 2000 IU Softgels Immune Support 260 Count   | vitamin d3, immune support, dietary supplements       | 94% 91% 87% |
| Natrol Melatonin Sleep Aid Gummies Strawberry 10mg 90 Count        | sleep aid, melatonin gummies, sleep supplements       | 93% 89% 82% |
| Vital Proteins Collagen Peptides Powder Vanilla 24 Oz              | collagen supplements, collagen powder, protein powder | 95% 93% 76% |
| Align Probiotic Daily Digestive Health 24 Hour Support 56 Capsules | probiotic supplements, digestive health, gut health   | 91% 88% 79% |

|                                                               |                                                                  |                                                         |                                                                   |
|---------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------|
| <b>Framework</b><br>TensorFlow 2.13 + Keras<br>Functional API | <b>Interface</b><br>Streamlit — Single + Batch<br>CSV + Insights | <b>Training</b><br>Google Colab T4 GPU  <br>~25 minutes | <b>No Overfitting</b><br>Val AUC ≥ Train AUC  <br>Dropout working |
|---------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------|

GitHub Repository contains: full source code | preprocessing pipeline | Colab training notebook | Streamlit app | model weights